

How hard does the blanket pull when it opens?

A first-fidelity dynamic model of the Flarewing solar blanket unfolding: the loads it puts on the deployment mechanism, the loads inside the blanket, and which unknowns to measure first.

02	The problem and the model
04	Is the math right? Verification
05	What happens during deployment: live
07	What drives the peak load
10	What to measure first
11	The plan: two-week sprint, stakeholders, who does what
12	Backup: where the numbers come from

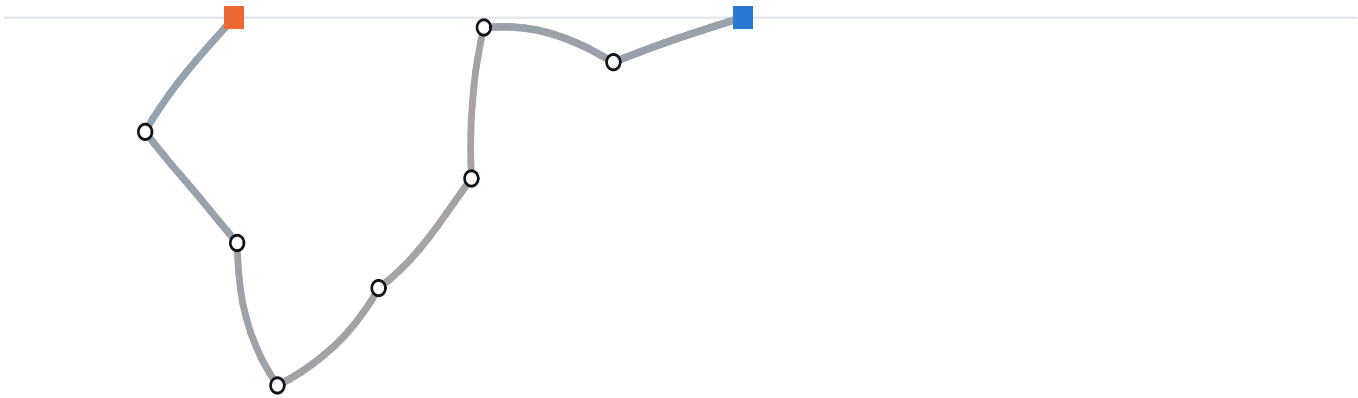


Fig. 1 Simulated baseline deployment at true scale: 8 × 0.5 m panels pulled open over 30 s, colored by tension (gray slack, red taut). Loops.

PROJECT Flarewing blanket deployment	SHEET 01 of 12	SUBJECT Title	STATUS Placeholder parameters · not validated	MODEL BlanketSim.jl
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Two questions, and no test data yet

- Q1

What loads does the blanket put on the mechanism?

Force at the root and tip attachments over time, especially the peak when the blanket snaps taut.
- Q2

What loads does the blanket take itself?

Tension along the film, fold angles, and cell curvature checked against the allowable bend radius of the silicon strips.

Cells are stiff along their strips and flexible across the gaps, so the blanket is modelled as **stiff panels joined by flexible fold lines** that remember their stowed shape. The mechanism itself is represented only by the motion it imposes.

Every material value is a placeholder. Today's job is **how loads scale** and **what to measure first**, not final numbers.

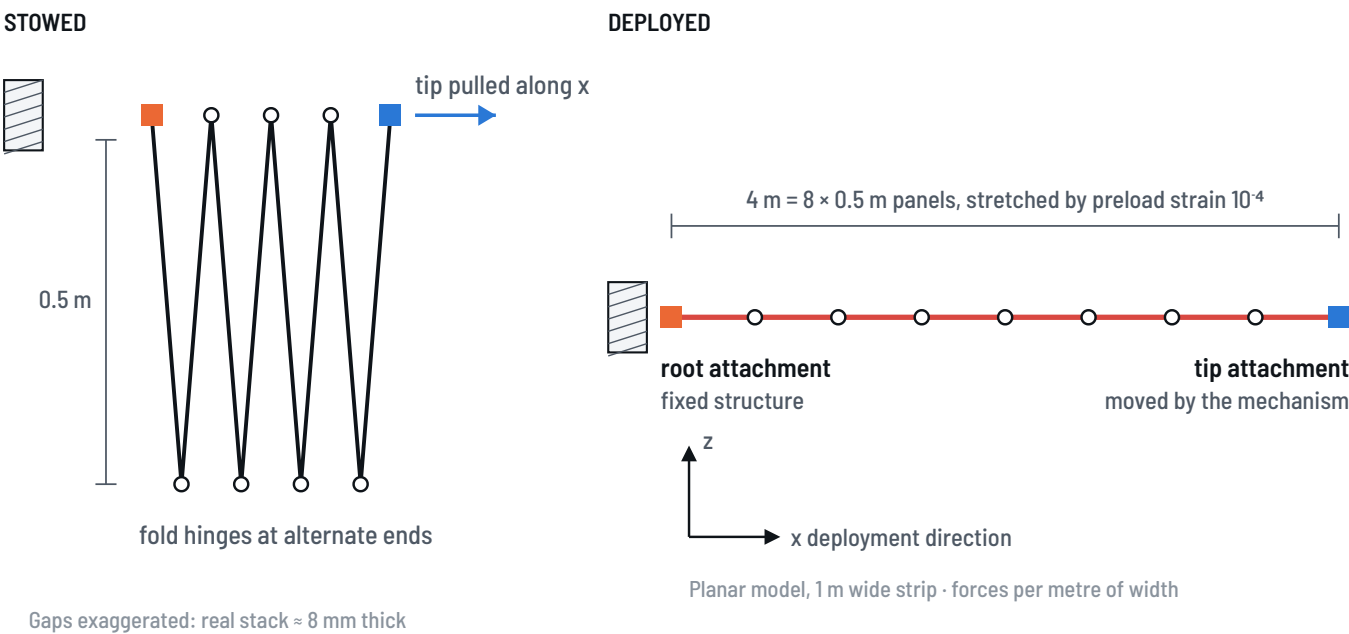


Fig. 2 The blanket is stowed as a z-fold stack and pulled open along x by the tip attachment until it is flat and slightly stretched. Stack thickness is exaggerated.

The blanket as beads and springs

Mass is lumped onto nodes every 25 mm. Springs between them carry stretching, bending and folding. The computer moves every node forward in 7.9 μs steps.

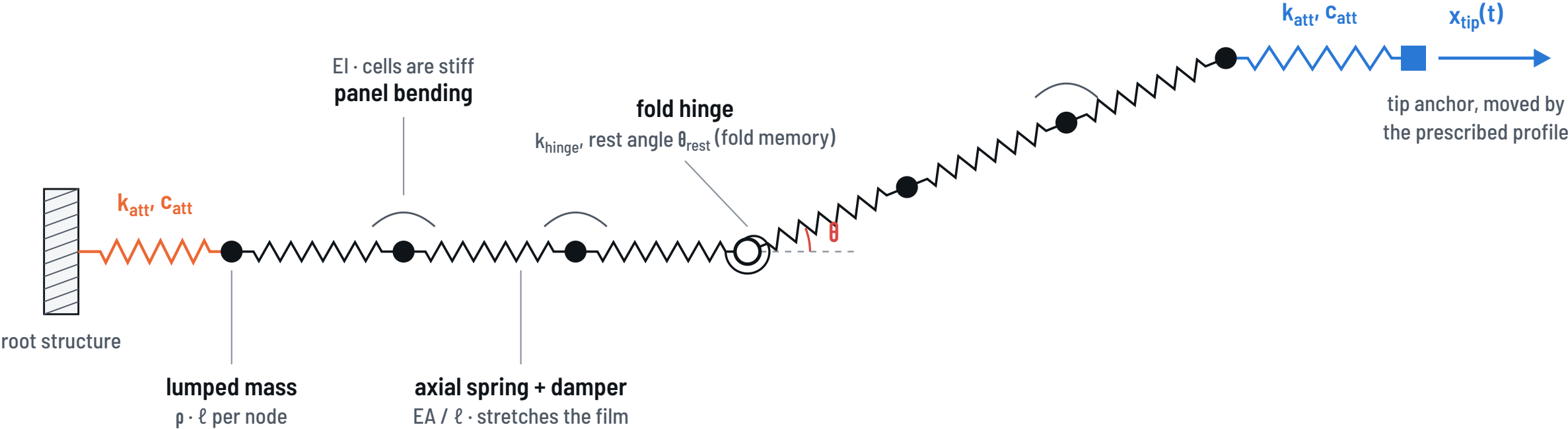


Fig. 3 One strip, 1 m wide, in the x-z plane. Forces are per metre of width. Optional gravity and air drag act on every node.

161 nodes

8 panels × 20 segments, fold hinges at shared nodes

4.8 M steps

per 38 s run, about 35 s of compute on one core

3 profiles

smooth S-curve, trapezoid, constant speed with an abrupt stop

10⁻⁷ balance

energy in = stored + kinetic + dissipated, checked every step

The numerics check out before the physics is trusted

CHECK AGAINST A KNOWN ANSWER	CRITERION	ERROR
Pendulum period: one stiff panel swinging under gravity	within 1%	0.08% ✓ pass
Fold hinge oscillation frequency	within 1%	0.34% ✓ pass
Energy conservation with no damping	drift < 0.5%	0.0004% ✓ pass
Energy balance during a driven deployment	closes within 1%	0.003% ✓ pass
Static tension equals $EA \times \text{strain}$	exact	exact ✓ pass
Axial wave speed equals $\sqrt{EA/\rho}$	within 5%	0.3% ✓ pass
Peak load: mesh doubled / time step halved	changes < 2%	0.43% / 0.006% ✓ pass
Hysteretic fold: loop work equals dissipation	within 1%	0.1% ✓ pass

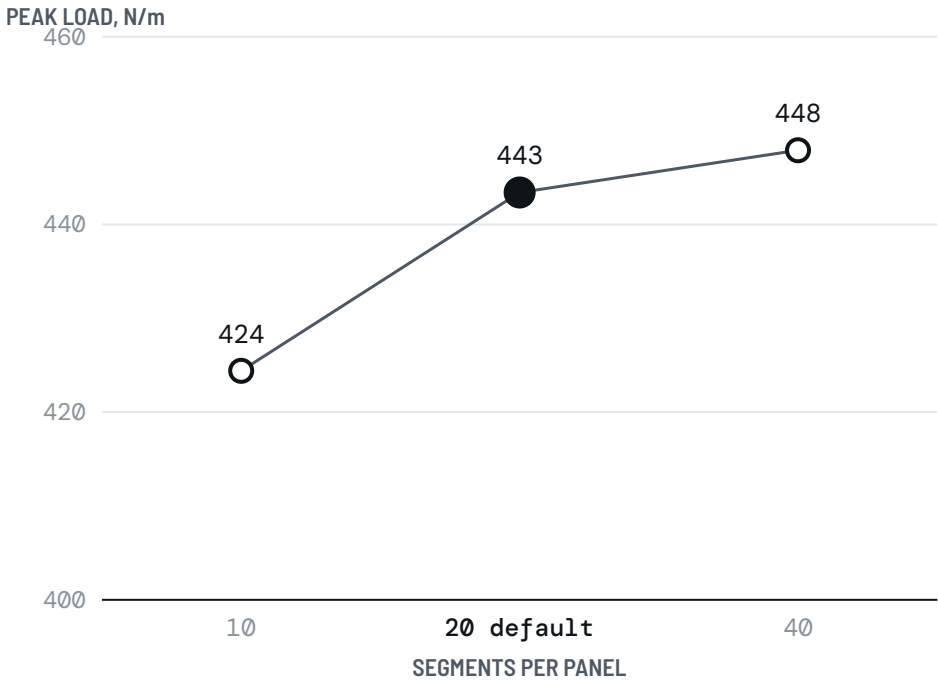


Fig. 4 Baseline peak attachment load at three mesh resolutions. 20 segments per panel (the default) is within 1% of 40. Halving the time step changes it by less than 0.01%.

In orbit, the peak load is a whip, not an impact

Baseline	Abrupt stop	Ground	Fold friction
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In orbit, smooth 30 s deployment. The blanket flaps after unfolding and whips taut near the end.

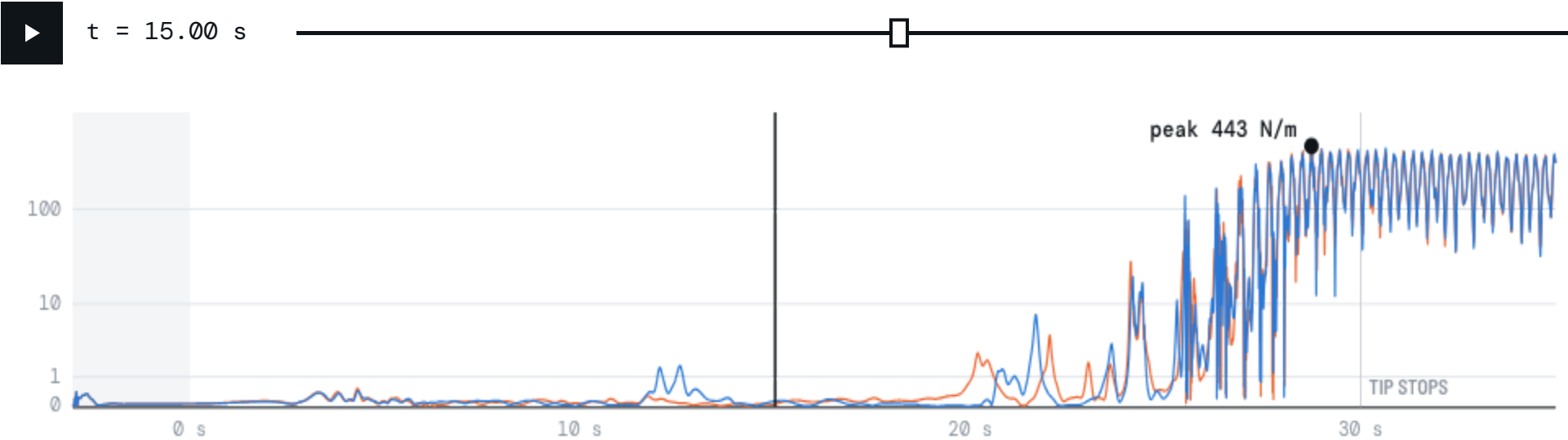
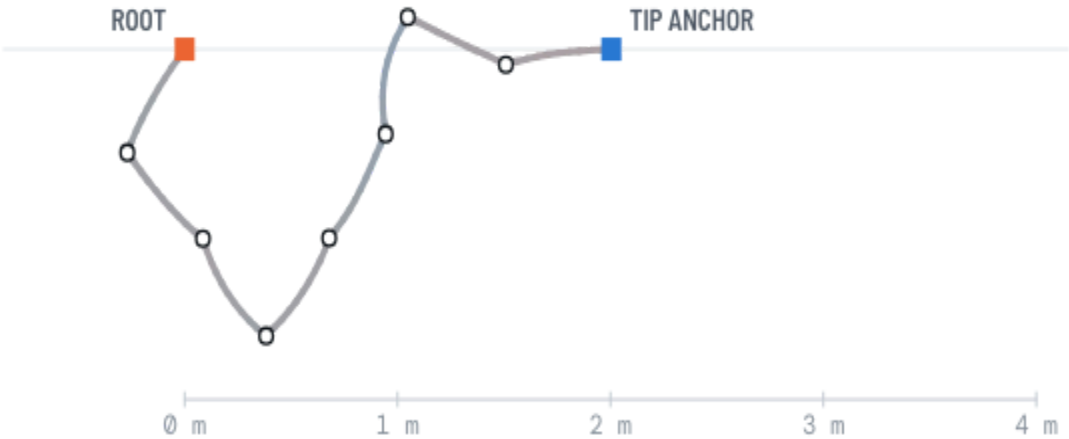
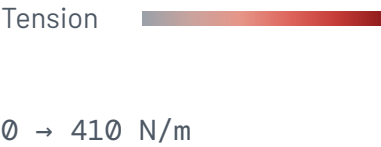


Fig. 5 Attachment loads per metre of width, log scale. Drag on the chart to scrub. ■ tip ■ root

Tip load	0.133 N/m
Root load	0.0556 N/m
Max tension	0.131 N/m
Cell curvature	1.03 × allow. OVER
Tip travel	50.0 %
Peak attachment load	443 N/m
vs preload (12.5 N/m)	×35
Peak cell curvature	3.3 × allow. OVER



Unfolding leaves it flapping; pulling it taut pumps the flapping up

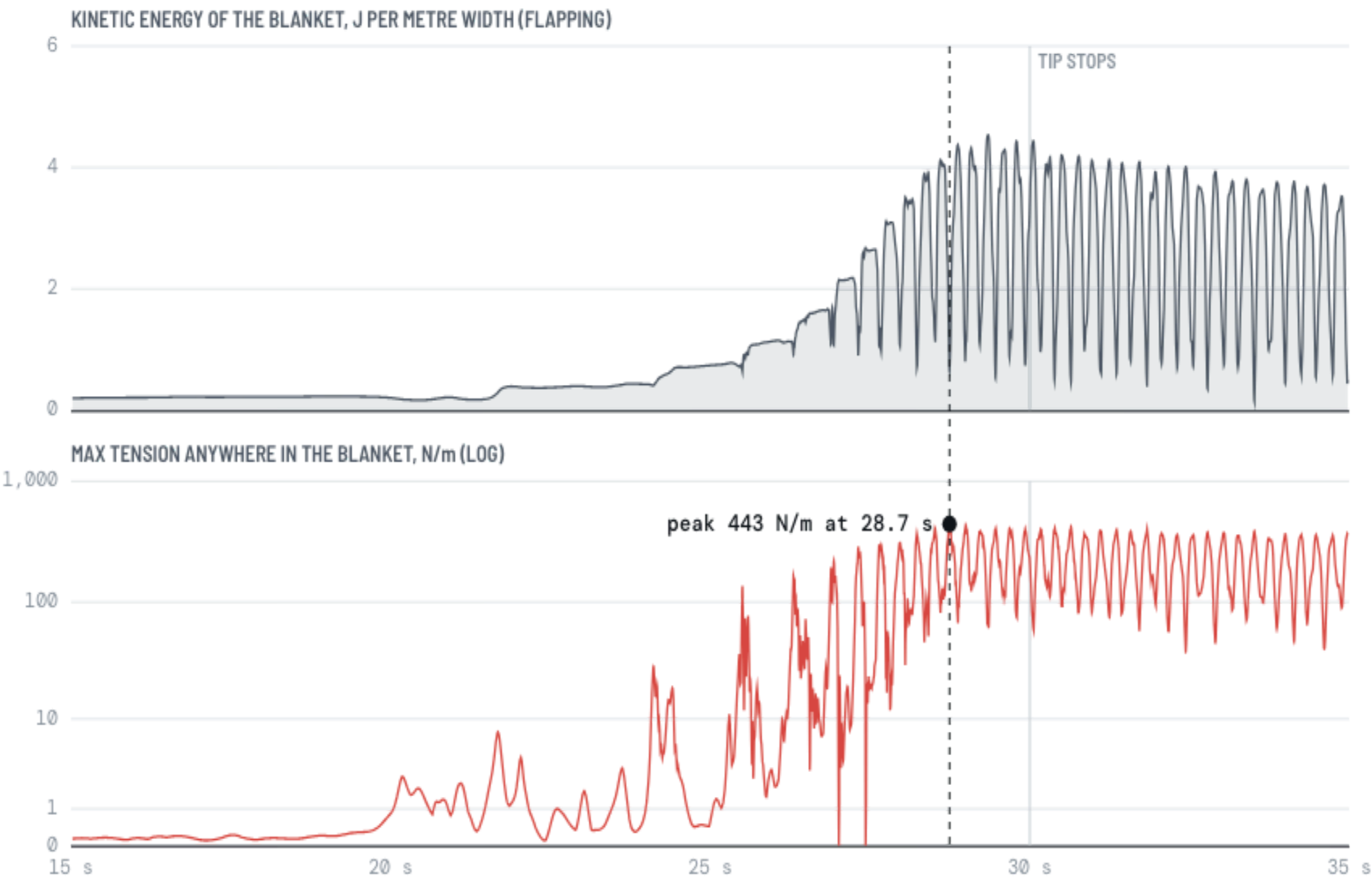


Fig. 6 Baseline, last 20 s. As the blanket is drawn taut its flapping gains energy until it snaps straight. Tension spikes at each snap while the tip anchor is barely moving.

×35

the preload tension: 443 N/m at the peak, against 12.5 N/m once it settles

6.7 mm/s

tip anchor speed at the peak: the mechanism is barely moving

≈ 40 ms

length of the peak, about 4× longer than an axial wave takes to cross the blanket (11 ms): not a wave impact

×9–10

drop in peak load from slowing deployment from 7.5 s to 120 s: less flapping energy to pump

Deployment speed and fold memory dominate the peak load



Peak load relative to the baseline run (432 N/m), log scale. Gray band ±4%: flapping-phase noise.

Smooth S-curve

Abrupt stop

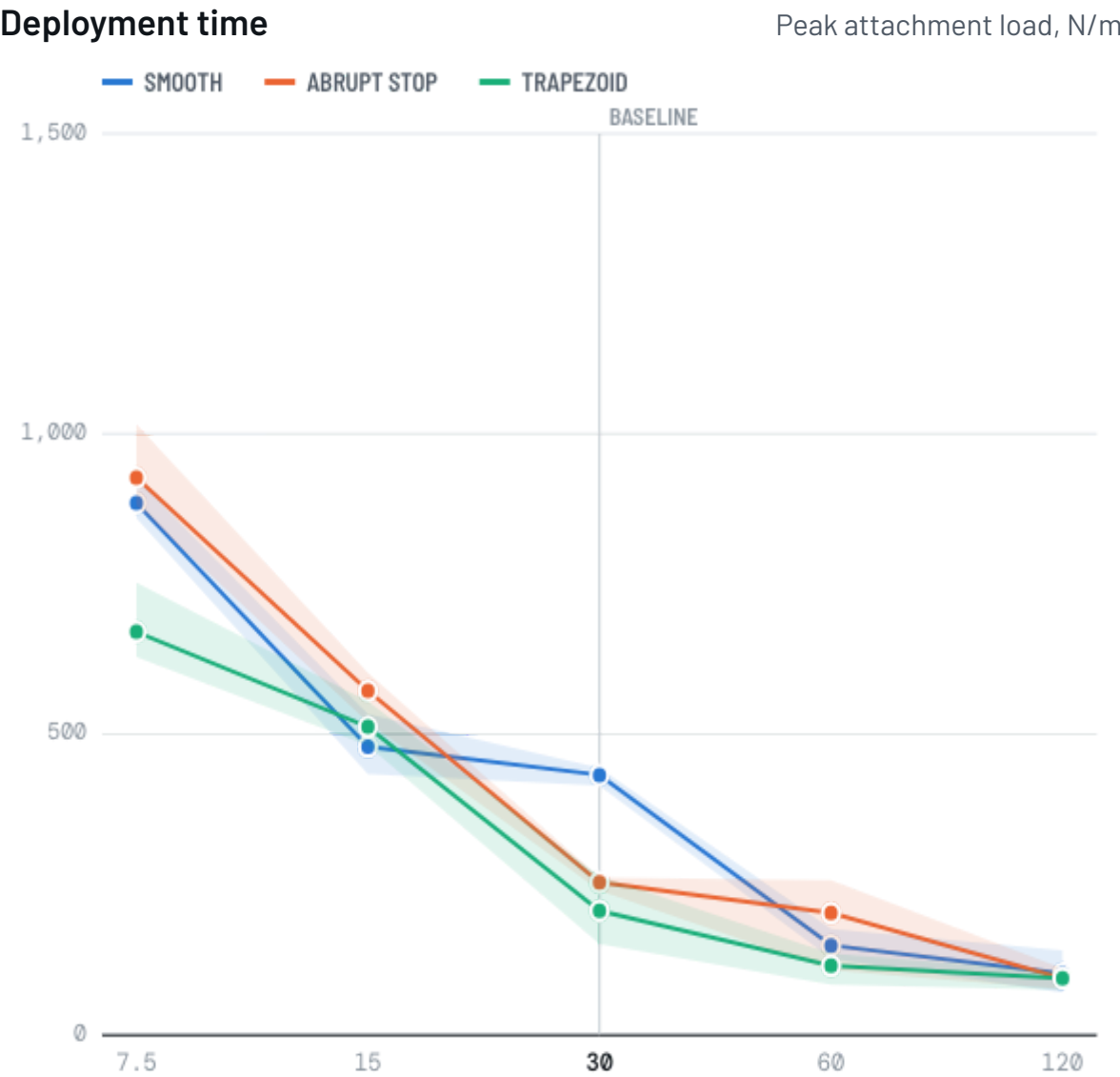


Fig. 7 One parameter varied at a time around the baseline. Bars and lines: mean of three runs with deployment time × 0.97, 1.00, 1.03; gray and shaded bands: their spread (flapping-phase noise).

A ground test in air will not see the in-orbit snap

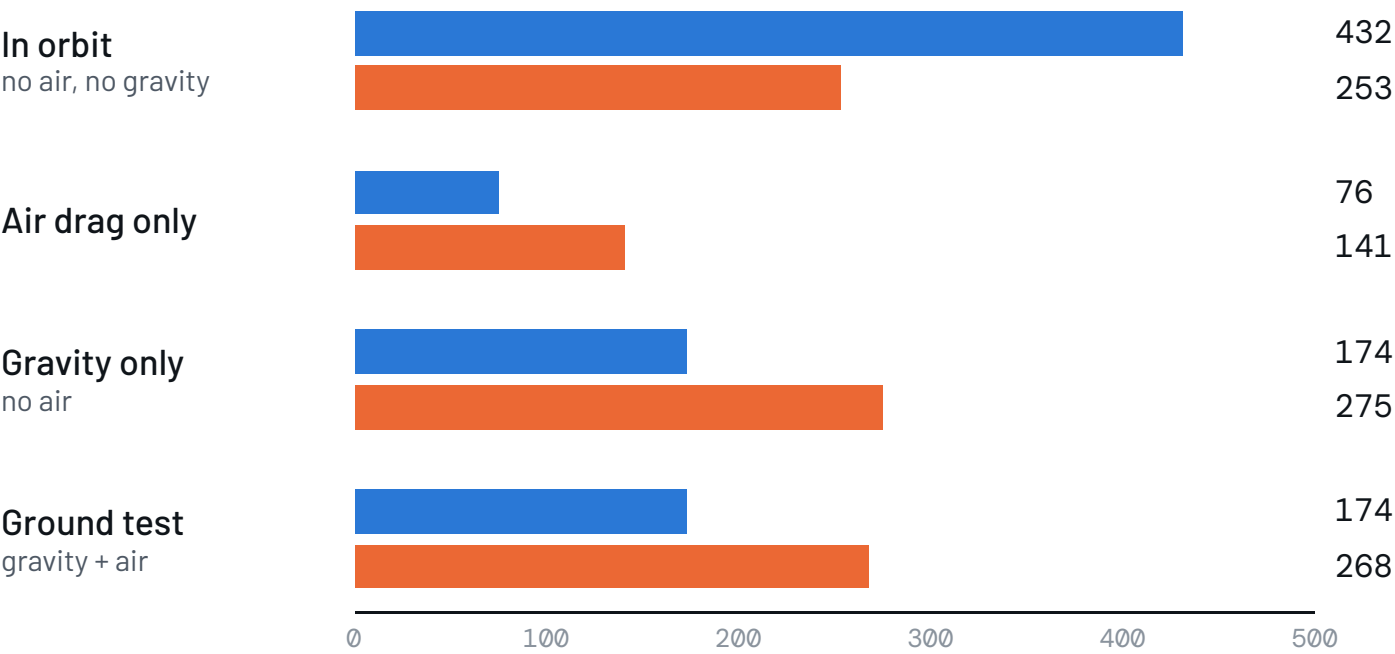


Fig. 8 Peak attachment load by test environment, N per metre of width. Mean of three runs each. ■ smooth S-curve ■ constant speed, abrupt stop

432 → 76 N/m

Air drag alone removes most of the in-orbit peak by damping the flapping that drives the whip.

≈ 174 N/m

With gravity, the ground load is simply the sag tension of the deployed blanket, not a dynamic peak.

To validate the in-orbit load case, plan at least one **vacuum or reduced-pressure** deployment and a **gravity offload**.

Every run bends the cells past the placeholder limit

Each fold springs back and pushes on the panel beside it. Moment over bending stiffness gives the curvature at the panel edge, before any dynamics:

$$\kappa \cdot R_{\min} \approx k_{\text{hinge}}(\pi - \theta_{\text{rest}}) \cdot R_{\min} / EI$$
$$= 0.026 \times 0.7\pi \times 0.5 / 0.01$$
$$\approx 2.9 \times \text{allowable} \text{ (3.3 simulated)}$$

$$EI \approx 0.03 \text{ N}\cdot\text{m}$$

Panel stiffness needed to stay within the allowable with placeholder fold properties: about 3× the placeholder.

Whether this is real depends on two unmeasured things: the fold moment-angle curve and panel EI.

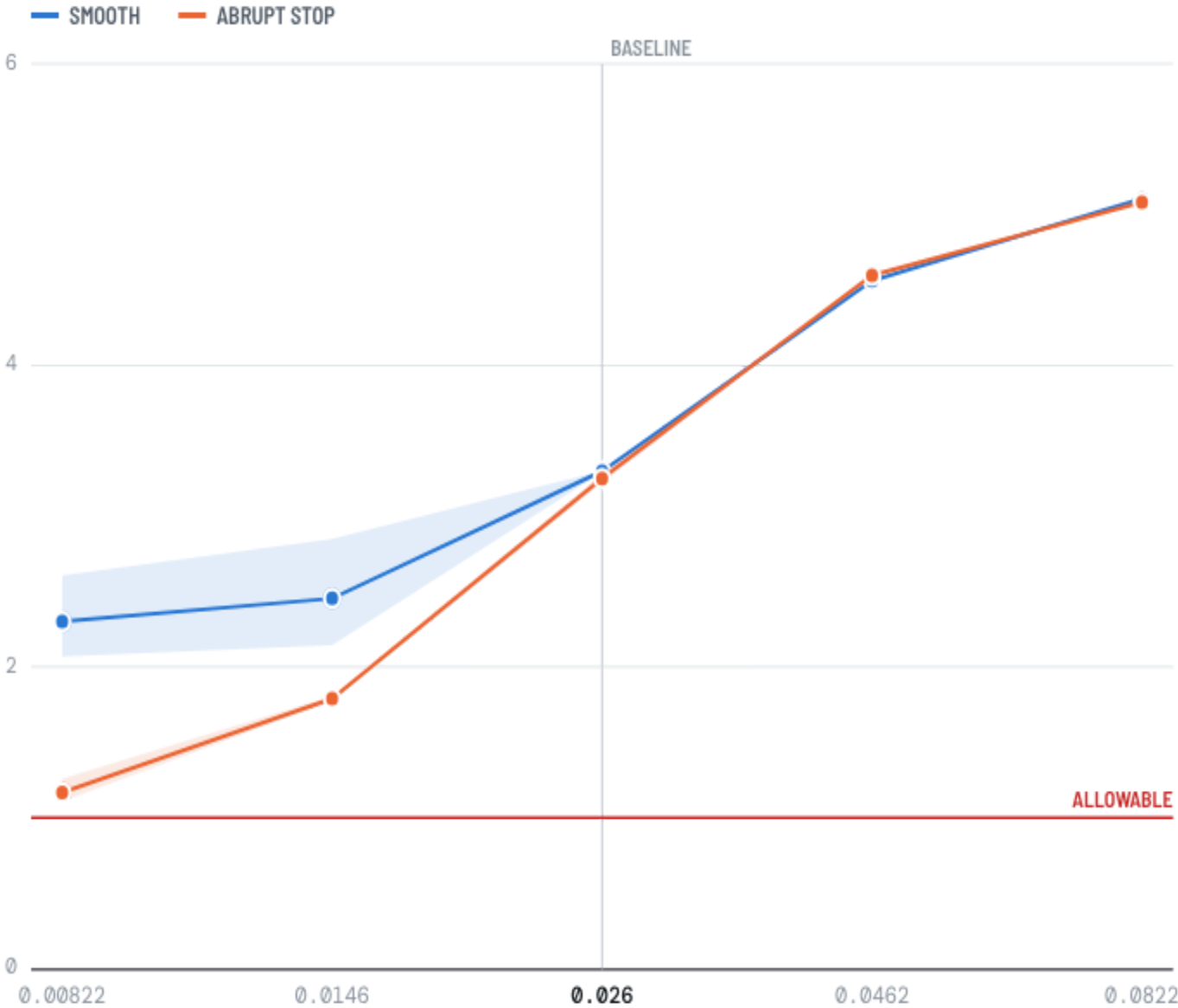


Fig. 9 Peak panel curvature relative to the allowable 1/R_{min} across the hinge-stiffness sweep. Stiffer folds bend the cells harder; red line = allowable.

What to measure first

	MEASURE	WHY IT MATTERS	EFFECT ON PEAK LOAD
1	Fold moment–angle curve at the stowed angle, with hysteresis, vs stowage time and temperature	Largest material driver of loads (stored springback energy) and the direct driver of cell curvature. A cheap coupon test that plugs straight into the model.	×2.6 – 6.7
2	Panel bending stiffness EI and the cell allowable $R_{\min}$	Decides whether the curvature exceedance is real.	cell check
3	One deployment in vacuum or reduced pressure, with gravity offload	Air removes most of the in-orbit whip; gravity replaces it with sag tension.	×5.7
4	Tip attachment and structure compliance	Monotone and predictable; also the easiest design lever.	×2.5
5	Damping from free-decay tests of folds and a partly deployed blanket	Sets how much flapping survives in orbit.	×1.5

DEPLOYMENT SPEED

The biggest lever of all (×9–10 between 7.5 s and 120 s) is a control choice, not an unknown. Exercise several speeds in every deployment test, with a load cell at the attachment.

LIMITS OF THIS MODEL

Planar strip; no contact between stack layers or stowage box; hinges are points; air is drag only. Curvature right beside a hinge is resolution-sensitive in fast snaps.

SOURCE

Code, figures, verification tests and the full sensitivity write-up: github.com/Rchari1/BRL_CaseStudy.

A two-week sprint to a first data-backed model

Two weeks gets real coupon data into the model and a first blind correlation. Work that needs facility time or long dwell tests follows.

	■ Days 1–3	□ Days 3–7	□ Days 8–10	□ After week 2
	1 · Kick off and plan PROTOTYPE BUILT	2 · Characterize the blanket COUPON TESTS IN AIR	3 · Deployment test BLIND PREDICTION	4 · Follow-ons AS RESOURCES ALLOW
MODEL	Prototype verified (58 checks); the sensitivity sweep picks which tests come first	Swap in measured fold curves (tabulated or hysteretic) and fitted damping	Issue predicted loads and fold angles before the test, then correlate	Gaps between cells, stack contact, air added mass; 3D model; coupling with the structure model; FE cross-check
TESTS AND DATA	Write the test plan and measurement requirements; build simple fixtures	Fold moment-angle with loading and unloading (short dwell); panel stiffness and cell bend limit; fold free-decay damping; attachment stiffness	Partial deployment or stow-and-release in air: load cell at the attachment, high-speed video, 2–3 speeds	Vacuum or reduced-pressure run with gravity offload; stowage-time and temperature dependence (needs long dwell)
DELIVERABLE	Test plan and prioritized unknowns (this briefing)	Measured parameter set and model v2	Correlation note, load estimates with uncertainty, updated test priorities	Design loads with margins, recommended deployment profile, analysis memo
STAKEHOLDERS	Mechanisms: drive profile, attachment design. Structures: attachment compliance	Test engineering: fixtures, instrumentation. Cell and blanket experts: bend limits, laminate data	Mechanisms: run the deployment. Systems: review results and next steps	Test facilities: chamber time, offload rig. Structures: coupled model, FE check

I WOULD OWN

- Building, verifying and maintaining the model and sweeps
- Test requirements: what to measure, sample rates, accuracy
- Data reduction, parameter fitting, blind predictions, correlation
- Load estimates, the analysis memo and recommendations

I WOULD ASK FOR HELP

- **Mechanisms:** real drive profiles, attachment design, which loads limit the mechanism
- **Structures analysts:** truss compliance, the coupled model, FE cross-check
- **Cell and blanket experts or supplier:** bend limits, laminate properties, how folds are made and stowed
- **Test engineering:** fixtures and instrumentation now; chamber time and gravity offload requested on day 1
- **Systems and leads:** load and deployment-time requirements, priorities

Where the numbers come from

SUPPLIED IN THE BRIEF, WITH THEIR PHYSICAL BASIS

PARAMETER	VALUE	BASIS	SWEPT
Film axial stiffness EA	1.25×10 ⁵ N/m	Kapton E ≈ 2.5 GPa × 50 μm thickness	–
Fold hinge stiffness k _{hinge}	0.026 N·m/rad	Film bending E·t ³ /12 ≈ 2.6×10 ⁻⁵ N·m, spread over a ~1 mm fold (k = EI/ℓ)	×0.3–3
Fold rest angle θ _{rest}	0.3π	Guess for fold memory (0 flat, π stowed)	0.1–0.95π
Panel bending stiffness EI	0.01 N·m	Cells + laminate, far stiffer than bare film	–
Areal density	1.0 kg/m ²	Rough total for cells + Kapton + encapsulant	–
Attachment stiffness	10 ⁵ N/m	Structure compliance at root and tip	10 ⁴ –10 ⁶
Damping ratios	0.05 / 0.02	Hinge / axial; placeholders, no basis given	hinge 0.01–0.2
Geometry	8 × 0.5 m	Placeholder panel count and length; 1 m wide strip	4–16 panels
Deployment time	30 s	Placeholder; three profile shapes	7.5–120 s
Preload strain	10 ⁻⁴	Final tension EA·ε = 12.5 N/m	–
Cell bend allowable R _{min}	0.5 m	Placeholder for the silicon strips	–
Air ρ, C _d · stow gap	1.2, 1.5 · 1 mm	Standard air, flat plate · layer gap	on / off

No test data exists, so every physical value is a placeholder with a stated basis. Nothing was tuned to make results look a certain way; each uncertain input was swept instead.

CHOSEN BY ME, BECAUSE THE MODEL NEEDED THEM

PARAMETER	VALUE	REASONING
Attachment damping	50 N·s/m	Small next to the blanket's own wave impedance $\rho c \approx 354$, so it does not dominate. Not swept.
Panel bending damping	$\zeta = 0.02$	Set equal to the axial ratio
Ratio → coefficient	per mode	Axial at the first mode of the deployed blanket; hinge against one panel swinging about it; bending at the first panel mode
Profile shapes	20% / 10%	Trapezoid acceleration fraction / abrupt-stop start ramp

NUMERICAL SETTINGS (NOT PHYSICS)

SETTING	VALUE	WHY
Mesh	20 per panel	Peak load within 1% of 40 per panel
Time step	7.9 μs	Half the computed stability limit; halving it changes peaks < 0.01%
Settle phase	3 s (8 s with g)	Stowed stack relaxes before t = 0; its damping switches off at t = 0

Because every input is uncertain, the deliverable is the **ranking**, not the number: verify the numerics on their own, sweep each placeholder over a decade or more, and let the sweep decide what to measure first.